

ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

Department of Software Engineering

DATABASE GROUP ASSIGNMENT

HOTEL MANAGEMENT SYSTEM

Using MySQL and MongoDB

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Chapter 1: Introduction

1.1 Background of the Organization

Addis Grand Hotel is a mid-sized three-star hotel located in Addis Ababa, Ethiopia. The hotel operates approximately 120 rooms categorized into four types: Single, Double, Deluxe, and Suite. It provides a full range of hospitality services including accommodation, room service, and restaurant facilities.

Currently, the hotel relies entirely on manual record-keeping and spreadsheets to manage its core operations — bookings, guest information, room status, and payment processing. As the hotel grows, this approach has become increasingly inadequate, prompting the need for a modern database-driven management system.

1.2 Problem Statement

The existing manual system at Addis Grand Hotel suffers from several critical limitations that directly impact operational efficiency and customer satisfaction:

- Frequent double bookings due to lack of real-time room availability tracking
- Manual errors in billing and payment records leading to financial discrepancies
- Loss or misplacement of guest records and historical data
- Inefficient handling of room service orders and guest feedback
- Difficulty in generating timely management reports for decision-making
- Increased staff workload and reduced service quality

These challenges collectively result in poor customer experience, revenue leakage, and operational inefficiency. A modern hybrid database system is essential to address these issues.

1.3 Objectives

General Objective

To design and implement a hybrid database system using MySQL (relational) and MongoDB (NoSQL) to automate and improve the overall efficiency of hotel management operations at Addis Grand Hotel.

Specific Objectives

- To analyze the hotel's data requirements and design a fully normalized relational database schema using MySQL
- To leverage MongoDB for storing flexible and semi-structured data such as guest reviews, feedback, and room service orders

- To enable real-time room availability checking and accurate booking management
- To automate payment tracking and generate management reports (occupancy rate, revenue, guest history)
- To improve data integrity, reduce manual errors, and enforce referential consistency

1.4 Scope and Limitations

The system is designed to cover the core operations of Addis Grand Hotel, specifically:

- Room type management and room availability tracking
- Guest registration and profile management
- Booking creation, modification, and check-in/check-out processing
- Payment recording and receipt generation
- Room service order management (via MongoDB)
- Guest reviews and feedback collection (via MongoDB)

The following are explicitly out of scope for this phase:

- Online payment gateway integration (e.g., PayPal, Stripe)
- Mobile application development
- Integration with third-party booking platforms (e.g., Booking.com)
- Multi-branch or multi-hotel support

1.5 Benefits of the System

Benefit Area	Description
No Double Bookings	Real-time room availability eliminates reservation conflicts
Faster Check-in/out	Automated guest lookup and billing reduces processing time
Flexible Data Handling	MongoDB's schema handles varied guest reviews and service orders efficiently
Financial Accuracy	Automated payment tracking removes billing errors and revenue leakage
Better Reporting	Instant generation of occupancy, revenue, and guest history reports
Improved Experience	Faster service and accurate records lead to higher guest satisfaction

1.6 Methodology and Tools

The project followed the Waterfall software development methodology, progressing through requirements analysis, system design, implementation, testing, and documentation phases in sequence.

Tools and Technologies Used

Tool	Purpose	Category
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MySQL 8.0	Structured transactional data storage	Relational Database
MongoDB	Document-oriented & semi-structured data	NoSQL Database
MySQL Workbench	Schema design and query execution	IDE / GUI
MongoDB Compass	Collection management and query testing	IDE / GUI
draw.io	ER Diagram design	Design Tool
GitHub	Version control and collaboration	DevOps

Chapter 2: Database Design

2.1 Entities and Relationships

The following core entities were identified through requirements analysis with hotel staff:

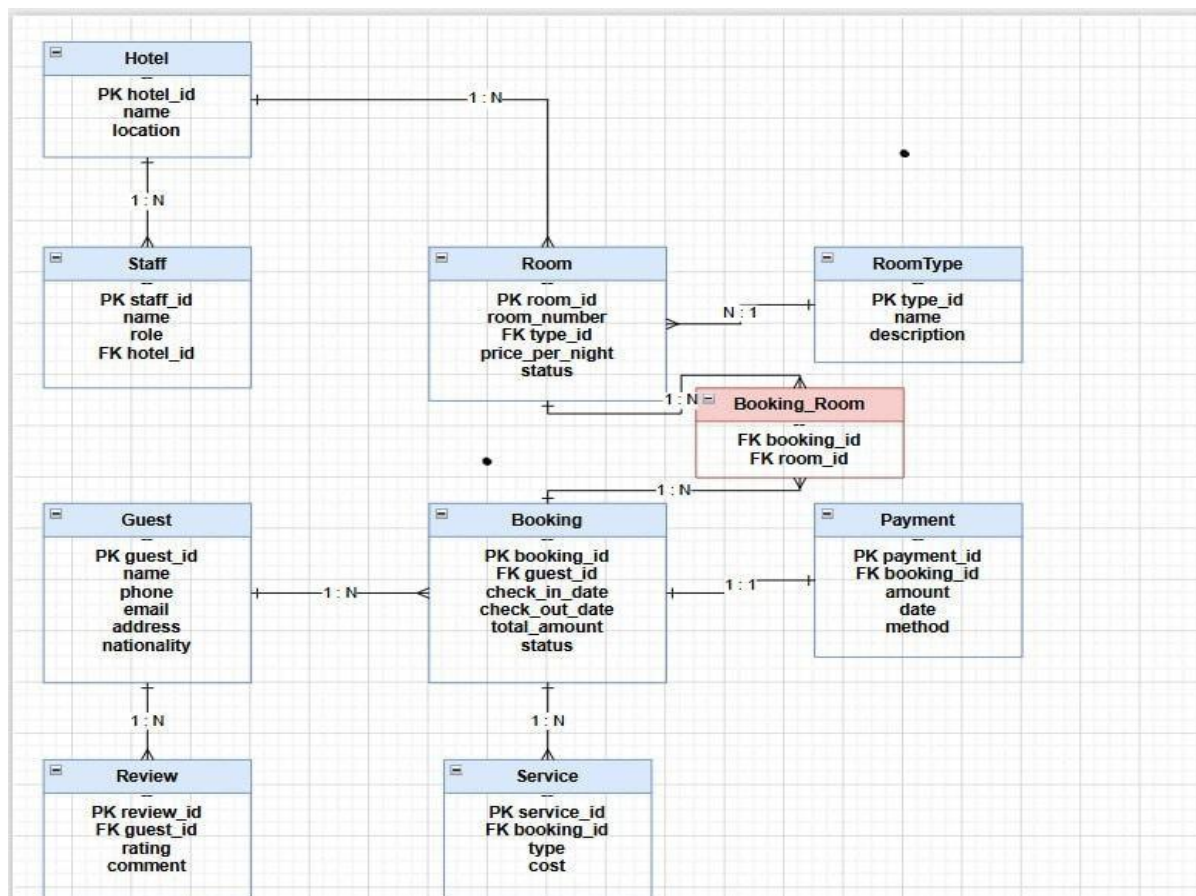
Entity	Database	Description
RoomType	MySQL	Categorizes rooms (Single, Double, Deluxe, Suite) with pricing
Room	MySQL	Individual room records with floor number and availability status
Guest	MySQL	Guest personal data, contact information, and ID details
Booking	MySQL	Reservation records linking guests to rooms with dates and status
BookingRoom	MySQL	Junction table resolving M:M relationship between Booking and Room
Payment	MySQL	Payment records linked to bookings with method and amount
Review	MongoDB	Flexible guest reviews with ratings, comments, and tags
RoomServiceOrder	MongoDB	Room service orders with variable item arrays and status

Key Relationships

- Guest → Booking: One-to-Many (1:M) — A guest can make multiple bookings
- Booking ↔ Room: Many-to-Many (M:M) — Resolved via the BookingRoom junction table
- Room → RoomType: Many-to-One (M:1) — Each room belongs to exactly one room type
- Booking → Payment: One-to-Many (1:M) — A booking may have multiple payment installments
- Guest → Review (MongoDB): One-to-Many — Guest IDs are referenced in MongoDB documents
- Booking → RoomServiceOrder (MongoDB): One-to-Many — Orders reference booking_id

2.2 ER Diagram

The Entity-Relationship (ER) diagram below illustrates all entities, their attributes, primary/foreign keys, and the cardinality of each relationship. The diagram was created using draw.io.



2.3 Logical Schema

The relational schema was implemented in MySQL with the following table structures:

Table	Attributes (PK underlined, FK noted)
RoomType	<u>type_id</u> (PK), type_name, price_per_night, max_occupancy, description
Room	<u>room_id</u> (PK), room_number, type_id (FK → RoomType), status, floor_number
Guest	<u>guest_id</u> (PK), full_name, phone, email, address, nationality, id_number
Booking	<u>booking_id</u> (PK), guest_id (FK → Guest), check_in_date, check_out_date, total_amount, status
BookingRoom	booking_id (FK → Booking), room_id (FK → Room) — Composite PK
Payment	<u>payment_id</u> (PK), booking_id (FK → Booking), amount, payment_date, payment_method

2.4 Normalization (up to BCNF)

The database was normalized through all normal forms up to Boyce-Codd Normal Form (BCNF) to eliminate redundancy and enforce data integrity.

Form	Requirement	Action Taken
1NF	Atomic values; no repeating groups	All attributes hold single values; no multi-valued fields
2NF	No partial dependencies on composite keys	Extracted RoomType data from Room; created BookingRoom junction table
3NF	No transitive dependencies	price_per_night moved to RoomType (not Room); removed derived attributes
BCNF	Every determinant is a candidate key	Verified all functional dependencies; all determinants are candidate keys

2.5 Input Forms and Output Reports

Input Forms

The following data entry forms support hotel operations:

- Guest Registration Form — Captures personal details and ID information
- Room Booking Form — Records reservation dates, room type, and special requests
- Check-in / Check-out Form — Manages room assignment and departure
- Room Service Order Form — Records items ordered, quantities, and pricing
- Payment Receipt Form — Confirms payment amount, method, and date

Key Output Reports

- Room Availability Report — Real-time view of vacant and occupied rooms
- Daily Revenue Report — Total revenue by day, broken down by room type
- Guest Booking History — Full reservation and payment history per guest
- Occupancy Rate Report — Percentage utilization over a given date range
- Average Review Score Report — Aggregated ratings from MongoDB reviews

Chapter 3: Implementation

3.1 MySQL Implementation

MySQL was selected for all structured, transactional data due to its ACID compliance, robust JOIN capabilities, and support for referential integrity through foreign key constraints.

Schema Highlights

- All tables use AUTO_INCREMENT primary keys for consistent record identification
- Foreign key constraints enforce referential integrity across all relationships
- CHECK constraints validate booking dates (`check_out_date > check_in_date`)
- ENUM types are used for room status ('available', 'occupied', 'maintenance') and payment method
- Indexes are applied on frequently queried columns (`guest_id`, `check_in_date`)

Sample SQL Queries Implemented

Query Purpose	SQL Operation Used
Check room availability for a date range	SELECT with NOT EXISTS subquery on BookingRoom
Calculate total revenue per room type	JOIN + GROUP BY + SUM(amount) across Payment and RoomType
Get guest booking history	INNER JOIN across Guest, Booking, BookingRoom, Room
Compute monthly occupancy rate	Aggregation with DATEDIFF and COUNT
Find top-spending guests	GROUP BY guest_id ORDER BY SUM(total_amount) DESC

Full schema: `mysql/schema.sql` | Sample queries: `mysql/queries.sql`

3.2 MongoDB Implementation

MongoDB was chosen to manage semi-structured and variable data that does not fit cleanly into a relational schema. Its document model is ideal for guest reviews (with arbitrary tag arrays) and room service orders (with variable item lists).

Collections

- `reviews` — Stores `guest_id`, `booking_id`, rating (1–5), comment text, and an array of tags
- `roomServiceOrders` — Stores `booking_id`, `room_number`, an items array (name, qty, price), `total_cost`, and status

- **guestFeedback** — Stores post-stay feedback with category ratings and open-ended responses

Sample MongoDB Operations

Operation	Method Used
Insert a guest review	<code>db.reviews.insertOne({ guest_id, rating, comment, tags })</code>
Find all reviews for a guest	<code>db.reviews.find({ guest_id: ObjectId('...') })</code>
Calculate average rating per room type	<code>db.reviews.aggregate([\$group, \$avg])</code>
Get all orders for a booking	<code>db.roomServiceOrders.find({ booking_id: '...' })</code>
Update order status to 'delivered'	<code>db.roomServiceOrders.updateOne({ \$set: { status: 'delivered' } })</code>

Sample documents: `mongodb/collections.json` | Sample queries: `mongodb/queries.js`

3.3 Testing and Results

Both databases were populated with realistic sample data and tested against all defined query requirements. Results confirmed the system performs as expected.

Test Case	Result	Notes
Book a room for existing guest	PASS	Booking created; room status updated to 'occupied'
Attempt to book an occupied room	PASS	System correctly rejects double booking
Check-out and mark room available	PASS	Room status returns to 'available' on check-out date
Generate monthly revenue report	PASS	Accurate SUM across all confirmed bookings
Insert guest review in MongoDB	PASS	Document stored with tags array intact
Aggregate average ratings	PASS	Correct average computed per room type via pipeline
Cross-reference booking_id (MySQL→MongoDB)	PARTIAL	Manual ID sync required; no automatic trigger yet

3.4 Challenges, Bugs & Improvements

Challenges Encountered

- Data synchronization between MySQL and MongoDB: `guest_id` and `booking_id` must be manually kept consistent across both systems, as there is no native cross-database referential integrity
- Schema design for normalization: determining the correct level of decomposition (especially for BCNF) required multiple iterations
- MySQL date handling: initial booking queries used incompatible date format strings, requiring format standardization

Bugs Resolved

- Date format mismatch in booking availability query — fixed by standardizing all date inputs to YYYY-MM-DD format
- Missing NOT NULL constraints on booking dates — added to prevent incomplete booking records
- MongoDB aggregation pipeline returning incorrect room averages — fixed by correctly grouping on `room_type` field

Recommended Future Improvements

1. Implement user authentication with role-based access control (admin, receptionist, manager)
2. Build a backend integration layer (Node.js or PHP) to orchestrate operations across both databases
3. Add real-time notifications for new bookings and check-out reminders
4. Introduce automated data backup and point-in-time recovery mechanisms
5. Develop a web-based frontend dashboard for management reporting
6. Integrate a payment gateway (e.g., Telebirr, CBE Birr) for digital payment processing

References

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- [4] C. J. Date, An Introduction to Database Systems, 8th ed. Boston, MA: Addison-Wesley, 2004.

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Annex A: Survey / Interview Questions

Project Title: Hotel Management System

Organization: Addis Grand Hotel, Addis Ababa

Interview Date: April 2026 | **Conducted by:** Group 7

Section 1: Current System & Challenges

7. How do you currently manage room reservations and bookings?
8. What are the biggest problems you face with the existing manual or spreadsheet-based system?
9. How do you handle guest check-in and check-out processes?
10. How do you track room availability and room status throughout the day?
11. What challenges do you encounter with billing and payment record-keeping?

Section 2: Guest Management

12. How do you currently store guest information (name, contact details, ID number, etc.)?
13. Do guests frequently make special requests? If so, how are these recorded and fulfilled?
14. How do you currently handle guest complaints or feedback after their stay?

Section 3: Room Service & Additional Services

15. How do you manage room service orders and restaurant orders placed by in-house guests?
16. How do you calculate and record payments for additional services beyond room charges?

Section 4: Reporting & Future Needs

17. What types of reports do you require on a daily or weekly basis (e.g., occupancy rate, revenue summary, guest list)?
18. What specific improvements would you like to see in a new digital management system?

Annex B: Sample Forms

Form 1: Guest Registration

Guest Full Name: _____
Phone Number: _____
Email Address: _____
Nationality: _____
ID / Passport Number: _____
Home Address: _____

Form 2: Room Booking

Guest Name: _____
Check-in Date: __ / __ / ____
Check-out Date: __ / __ / ____
Room Type: ☐ Single ☐ Double ☐ Deluxe ☐ Suite
Number of Guests: _____
Special Requests: _____

Form 3: Check-in / Check-out

Booking ID: _____
Guest Name: _____
Room Number: _____
Check-in Date & Time: ____ / ____ / ____ : ____
Check-out Date & Time: ____ / ____ / ____ : ____

Form 4: Room Service Order

Room Number: _____
Guest Name: _____
Order Date: __ / __ / ____

Item Description	Qty	Unit Price (ETB)	Total (ETB)
TOTAL AMOUNT (ETB):			

Special Instructions: _____

Form 5: Payment Receipt

Receipt No.: _____

Guest Name: _____

Booking ID: _____

Amount Paid (ETB): _____

Payment Method: ☐ Cash ☐ Debit/Credit Card ☐ Mobile Money (Telebirr / CBE Birr)

Date: ____ / ____ / ____

Received by: _____

Thank you for choosing Addis Grand Hotel
